Activity 5. Calculating versus estimating CEC

There are two ways you can calculate the CEC: the sum of cations method and the mineralogy method.

The Sum-of-Cations Method

If you have a soil analysis where the quantities of all cations in the soil are listed, simply summing all those exchangeable quantities will yield the CEC you found in the preceding problems.

The "Mineralogy" Method

As you know from your reading and class discussion, clay minerals have a range of values for CEC. If the mineralogy of the clay fraction is known (that is, the type and amounts of each clay mineral), then the CEC can be approximated.

To make these calculations easier, Table 13.4 contains representative values for CEC to use in all calculations for this class unless otherwise noted. In nature, however, these soil colloids will have a range of values.

Table 13.4. Typical CEC of various soil colloids.

Mineral or colloid type	**CEC of pure colloid**
	cmolc/kg
kaolinite	10
illite	30
montmorillonite/smectite	100
vermiculite	150
humus	200

As an example of this mineralogy approach to CEC calculations, consider a soil having 100% clay where the clay is 100% kaolinite. The CEC would then be 10 $cmol_c$/kg. If a soil contains only 10% kaolinite (or 10 kg clay in 100 kg soil), however, this clay would contribute

$$\text{Total CEC of the soil} = \frac{10\ \text{cmol}_\text{c}}{\text{kg clay}} \times \frac{10\ \text{kg clay}}{100\ \text{kg soil}} = \frac{1.0\ \text{cmol}_\text{c}}{\text{kg soil}}$$

A prairie soil contains 30% clay. This clay sized fraction is dominantly montmorillonite. The soil also contains 5% humus (organic matter).

Using the mineralogy method, what is the cation exchange capacity (CEC) contributed by the clay?